

DATA SCIENCE RESOURCES

Computer Vision Projects

Project 1. Face Mask Detection (Real-Time Compliance Monitor)

Problem Statement: In public spaces and hospitals, ensuring that people wear face masks is essential for public safety. Manual monitoring is inefficient.

Objective: Build a real-time face mask detection system using deep learning and OpenCV to detect whether a person is wearing a mask or not.

Dataset Source

- [Face Mask Detection Dataset](#) (Kaggle)
- [RMFD and IMFD Datasets](#)

Techniques & Concepts

- CNN for image classification
- Real-time video feed detection
- Face detection (Haar cascades, Dlib, MTCNN)
- MobileNetV2 transfer learning

Tools & Libraries: Python, TensorFlow/Keras, OpenCV, NumPy, Streamlit (for demo)

Expected Outcome: A live camera app that detects faces and classifies them into “Mask” or “No Mask” in real time.

Project 2. Road Lane Line Detection (Self-Driving Cars)

Problem Statement: Lane detection is crucial for autonomous driving systems to maintain vehicle trajectory and safety.

Objective: Build a computer vision pipeline that detects road lanes from dashcam videos in real time.

Dataset Source

- [Udacity Self-Driving Car Dataset](#)
- [KITTI Vision Benchmark](#)

Techniques & Concepts

- Canny edge detection
- Region of interest (ROI) masking
- Hough Line Transform
- Frame-by-frame lane overlay

Tools & Libraries: Python, OpenCV, NumPy, Matplotlib

Expected Outcome: A video or dashboard showing lane overlays in real-time or recorded driving footage.

Project 3. Pneumonia Detection from Chest X-rays

Problem Statement: Detecting pneumonia from X-ray images is time-consuming and requires experienced radiologists. AI can help automate early diagnosis.

Objective: Train a CNN model to classify X-ray images into “Pneumonia” and “Normal” using transfer learning.

Dataset Source: [Chest X-ray Pneumonia Dataset](#) (Kaggle)

Techniques & Concepts

- Image classification
- Transfer learning with ResNet or VGG
- Image augmentation
- Evaluation: accuracy, AUC, recall

Tools & Libraries: Python, Keras/TensorFlow, Matplotlib, OpenCV

Expected Outcome: A classification model that identifies pneumonia in X-ray images with high accuracy and can be deployed in medical screening apps.

Project 4. Real-Time Face Emotion Recognition

Problem Statement: Emotion recognition is critical for applications in e-learning, mental health monitoring, and human-computer interaction.

Objective: Detect and classify human emotions from facial expressions in real-time video feeds.

Dataset Source

- [FER2013 Emotion Dataset](#)
- [Extended Cohn-Kanade Dataset](#) (CK+)

Techniques & Concepts

- CNN for facial expression recognition

- Real-time face detection using Haar cascades
- Multiclass classification (Happy, Sad, Angry, etc.)
- Data augmentation for expression generalization

Tools & Libraries: Python, TensorFlow/Keras, OpenCV, Dlib, Matplotlib

Expected Outcome: A web camera-based application that labels emotional states (e.g., happy, angry, sad) with accuracy and real-time feedback.
